

CSE340: Computer Architecture
Assignment -4

Deadline: 10th May, 11:59 PM

Assignment must be handwritten. Scan and upload PDF in provided Google form

Question - 1:

Why does the ALU control unit take the following bits as input?

Instruction[30, 14-12]

Question - 2:

What is the significance of the Zero pin in the ALU?

Question - 3:

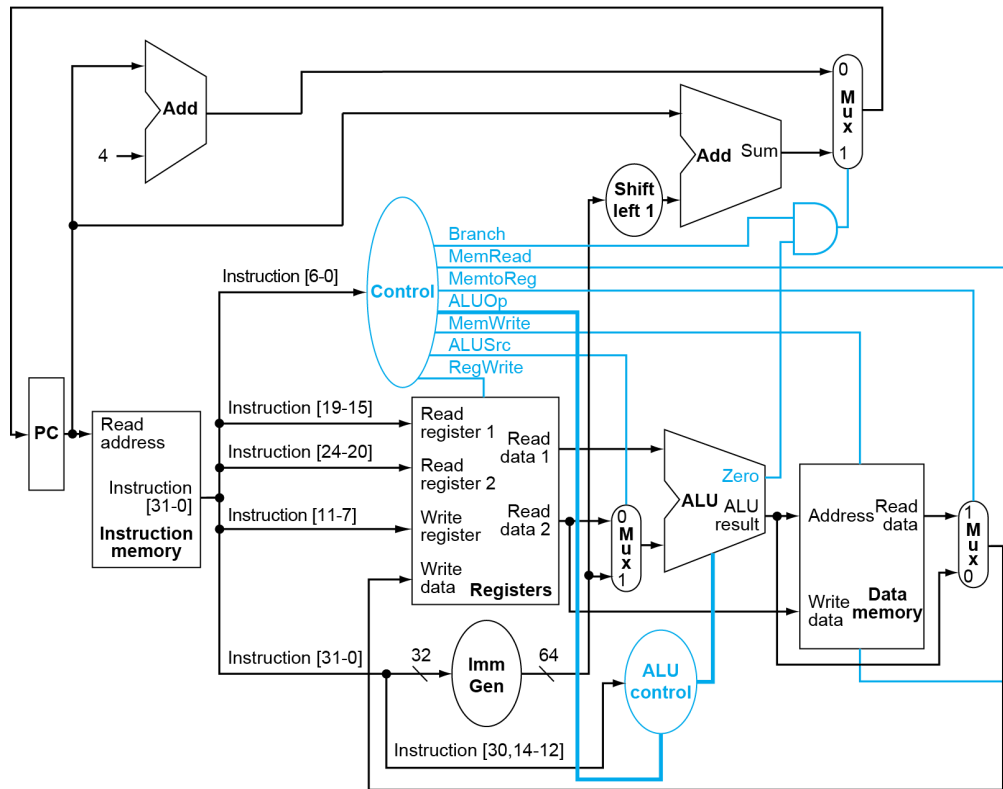
Suppose you have a single instruction:

Add X21, X22, X23

Register read	Instruction Fetch	Memory Operation	Register write	ALU Operation
20s	20s	40s	20s	40s

If you have a pipelined setup and a single-cycle setup,
Which one will run the instruction faster?

Question - 4:



Write the control signals for the following instruction:

Ld X5, 10(X21)

- i. ALU Src;
- ii. RegWrite;
- iii. MemRead;
- iv. MemWrite;
- v. MemtoReg

UNGRADED Problems (Only for Practice, no need to submit with Assignment)

1. Modify the BEQ datapath so that it works for BNE.
2. Explain control hazard in your own words.
3. Why do we keep instruction and data memory separate?
4. Write the control signals for the following instruction:

Addi X5, X6, 20

- a. i. ALU Src;
- b. ii. RegWrite;
- c. iii. MemRead;
- d. iv. MemWrite;
- e. v. MemtoReg